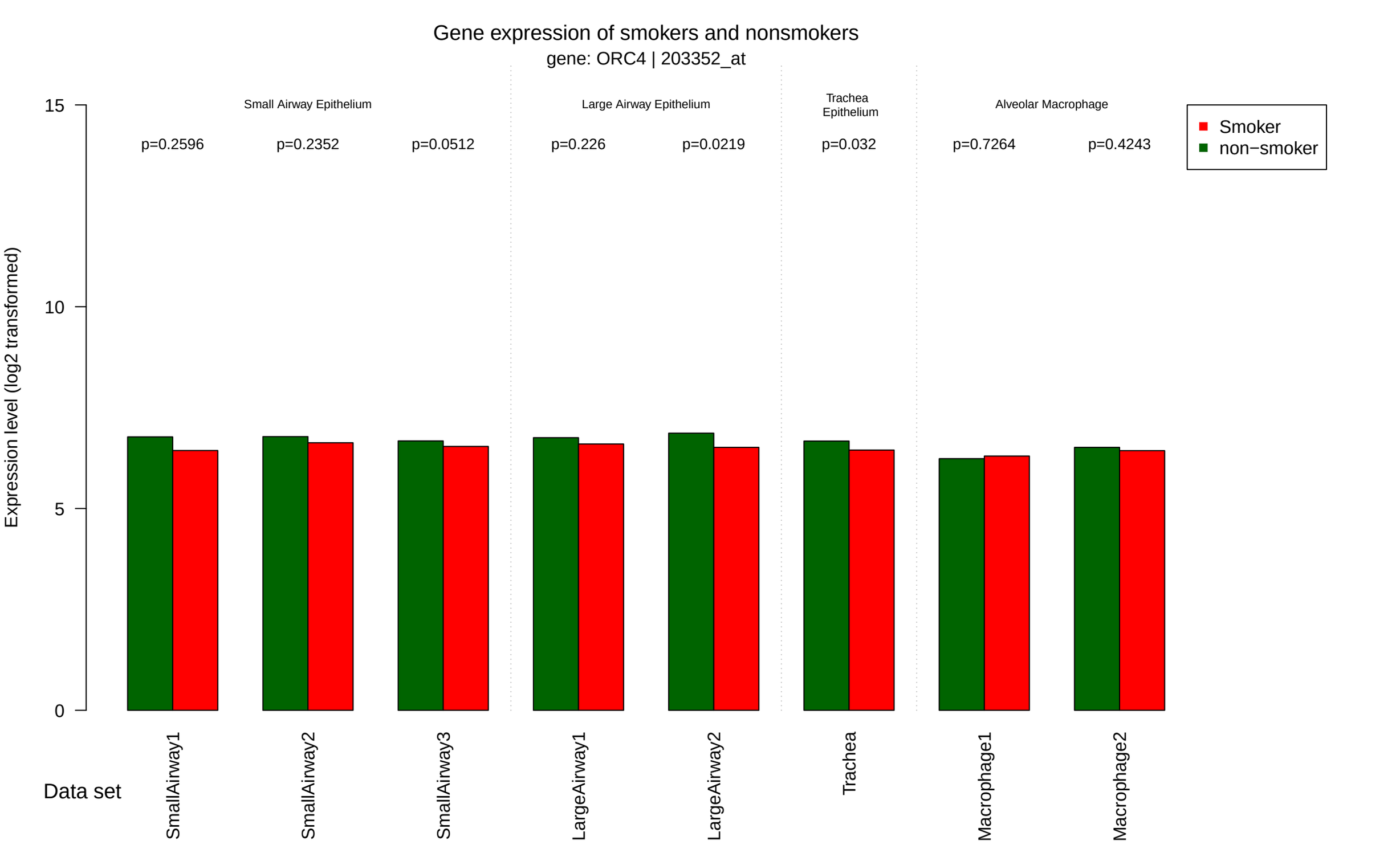
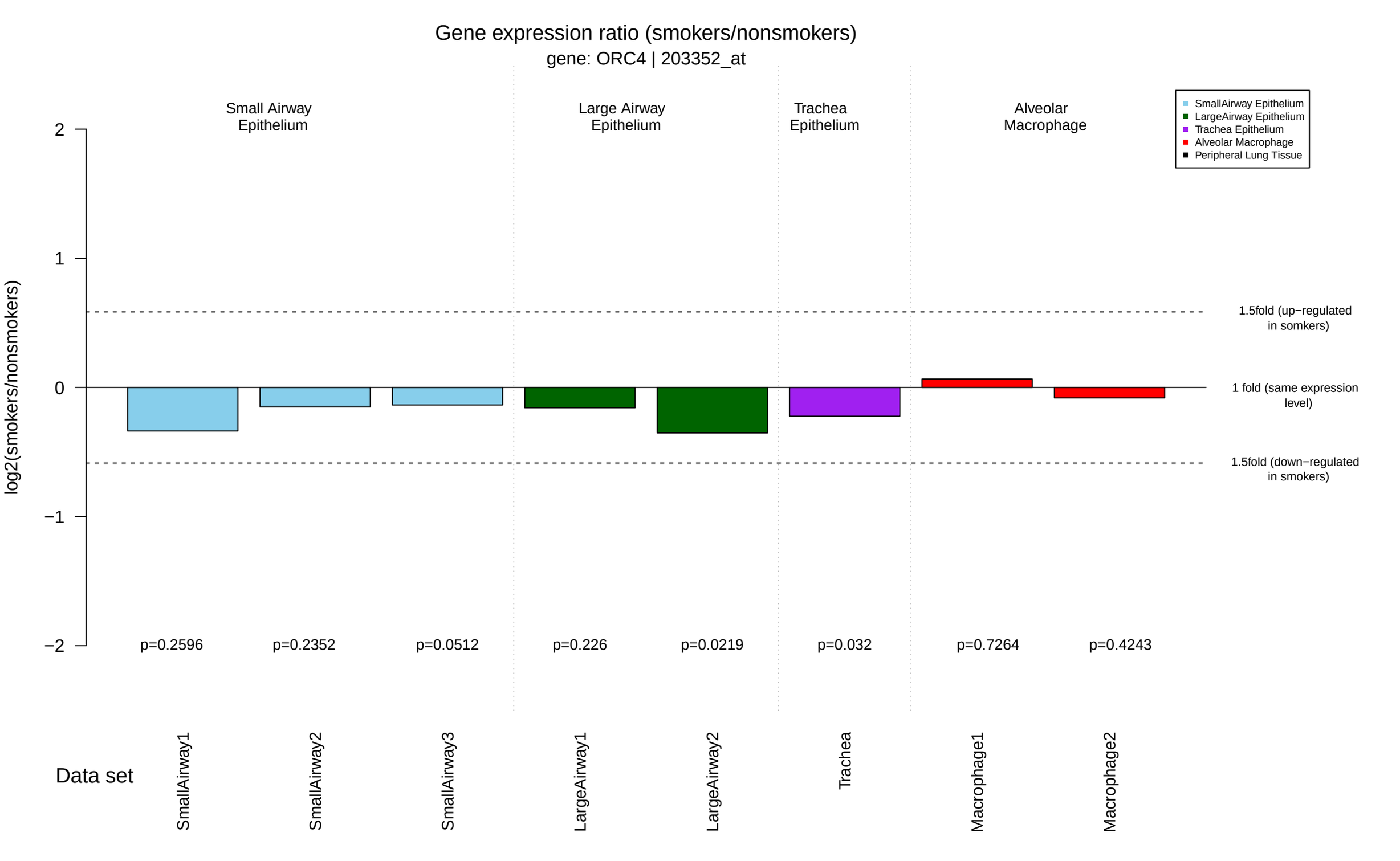
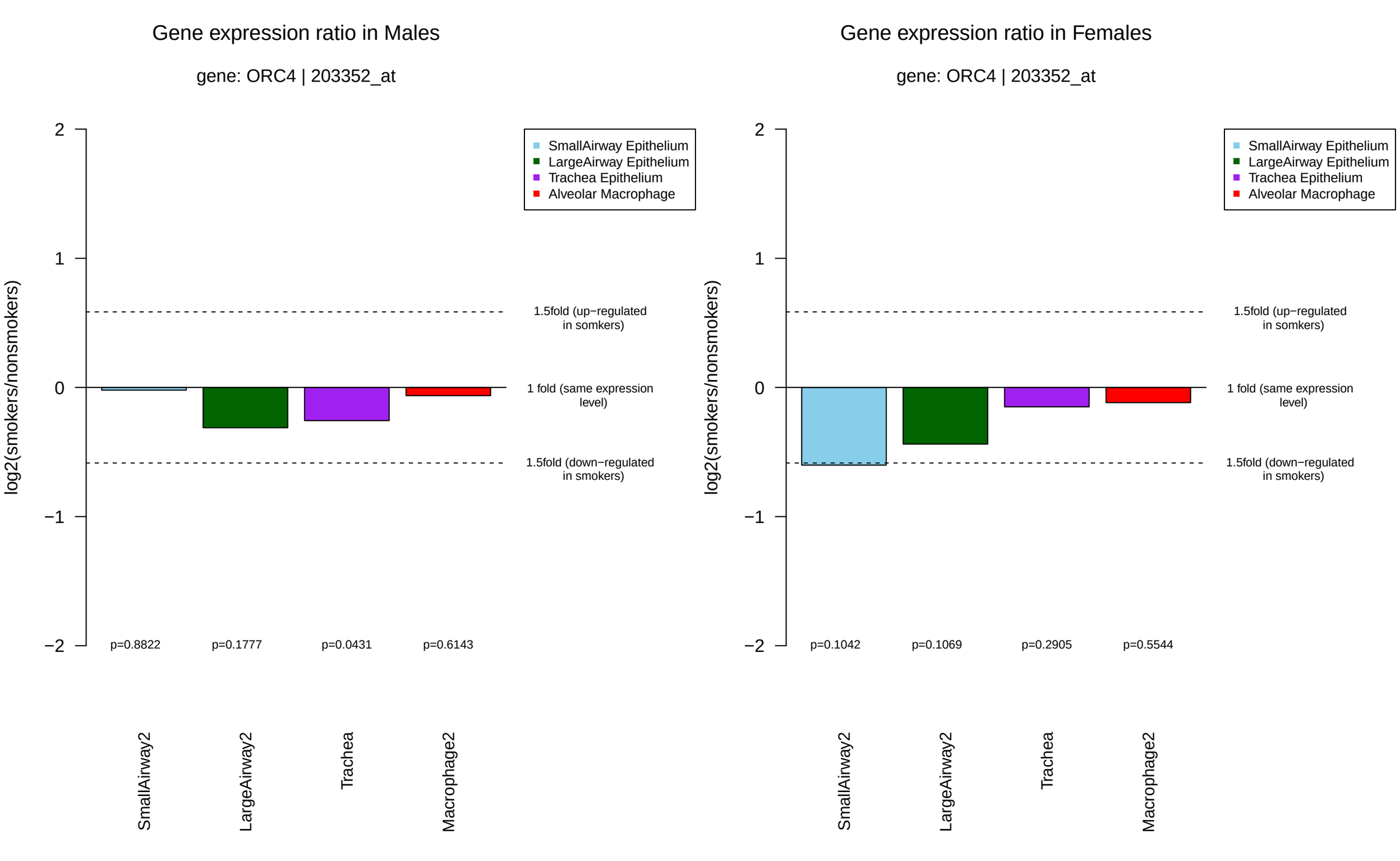
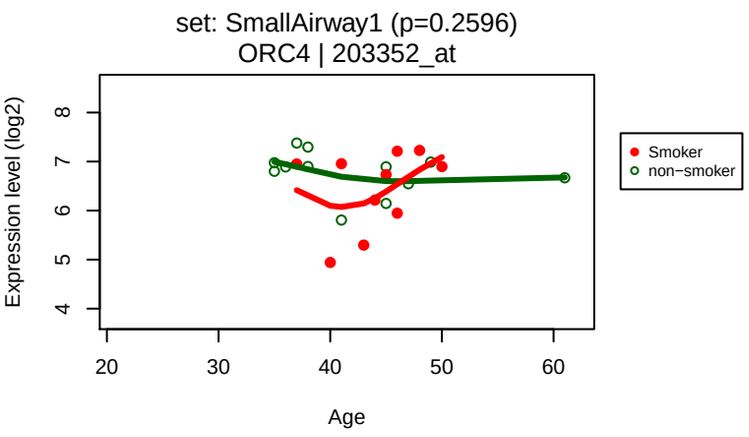
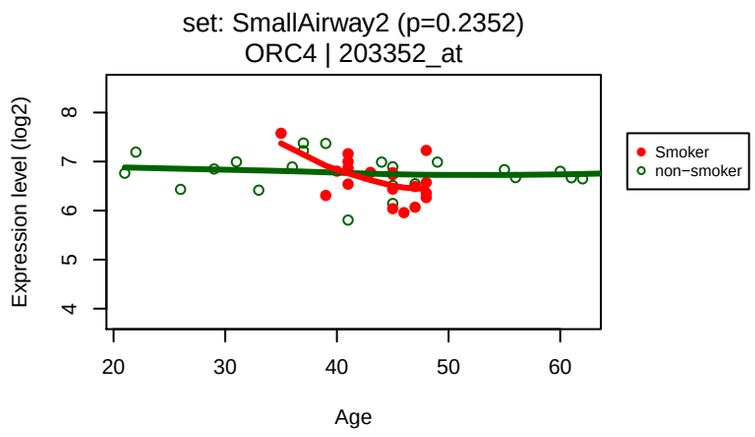
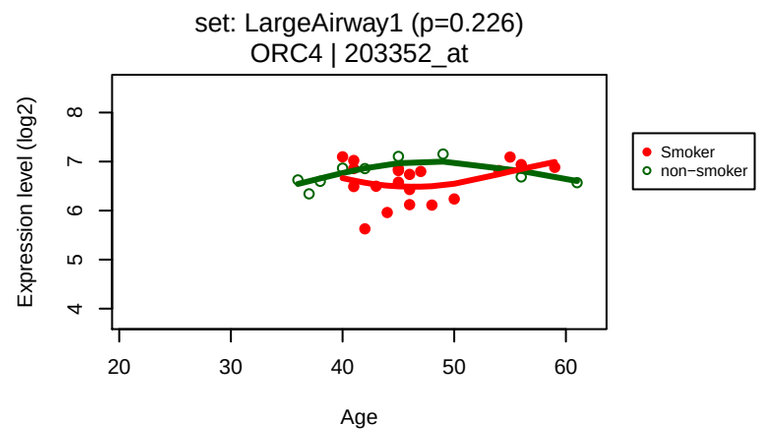

Gene expression of smokers and nonsmokers
gene: ORC4 | 203352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2596
p=0.2352
p=0.0512
p=0.226
p=0.0219
p=0.032
p=0.7264
p=0.4243
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

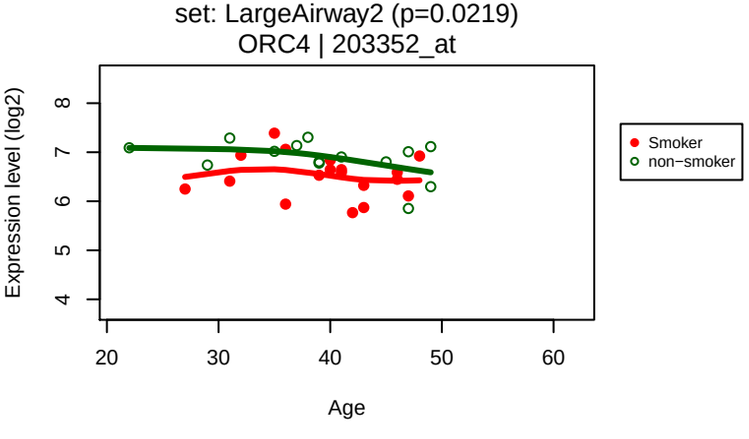

Gene expression ratio (smokers/nonsmokers)
gene: ORC4 | 203352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2596
p=0.2352
p=0.0512
p=0.226
p=0.0219
p=0.032
p=0.7264
p=0.4243
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

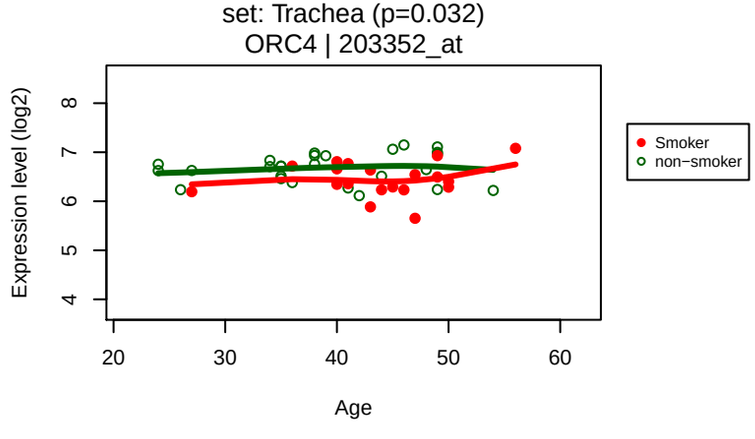

Gene expression ratio in Males
gene: ORC4 | 203352_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8822
p=0.1777
p=0.0431
p=0.6143
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ORC4 | 203352_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1042
p=0.1069
p=0.2905
p=0.5544
SmallAirway2
LargeAirway2
Trachea
Macrophage2

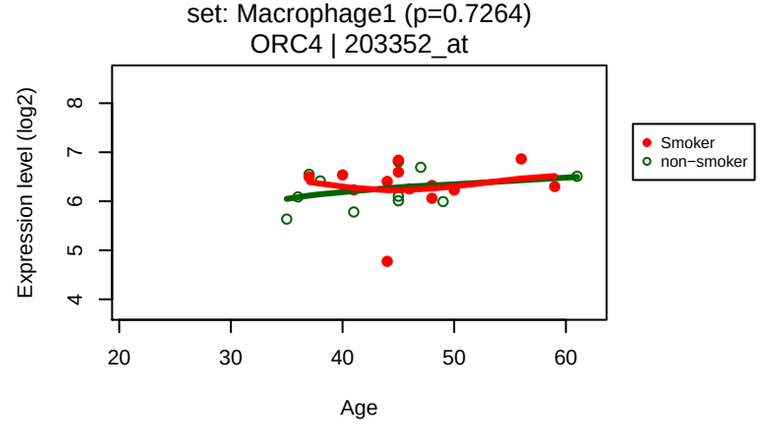

set: SmallAirway1 (p=0.2596)
ORC4 | 203352_at
Expression level (log2)
Age
Smoker
non−smoker

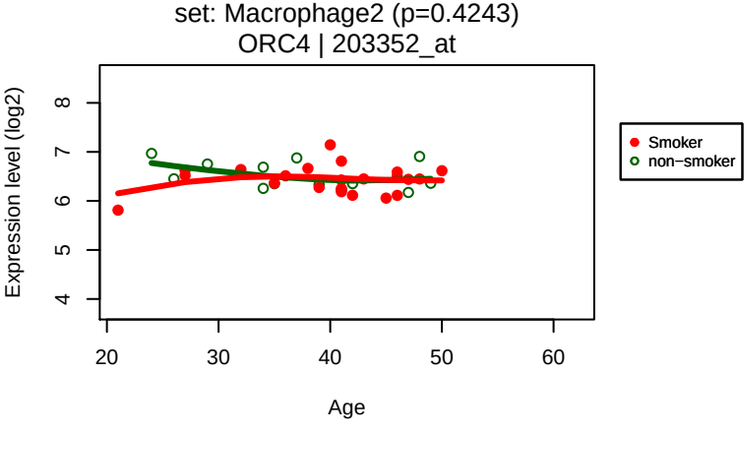

set: SmallAirway2 (p=0.2352)
ORC4 | 203352_at
Expression level (log2)
Age
Smoker
non−smoker

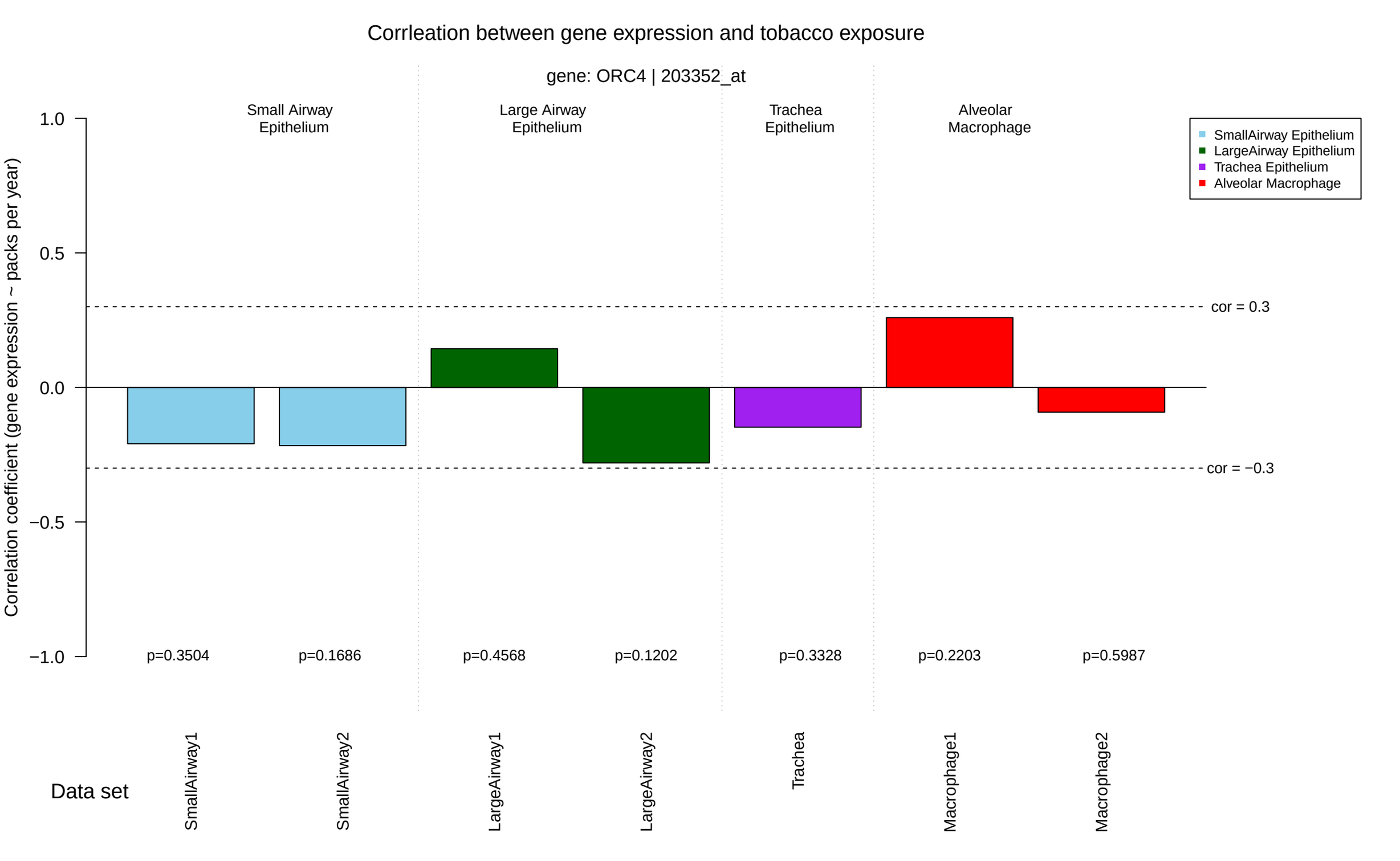

set: LargeAirway1 (p=0.226)
ORC4 | 203352_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0219)
ORC4 | 203352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.032)
ORC4 | 203352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7264)
ORC4 | 203352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4243)
ORC4 | 203352_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ORC4 | 203352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3504
p=0.1686
p=0.4568
p=0.1202
p=0.3328
p=0.2203
p=0.5987
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2